

Hengrong Du

CONTACT INFORMATION	Fisk University Department of Math & Computer Science	Page: https://hengrongdu.netlify.app/ E-mail: hdu@fisk.edu
RESEARCH INTERESTS	Partial Differential Equations for Machine Learning and Complex Fluids	
ACADEMIC POSITIONS	Fisk University , Nashville, TN, USA Assistant Professor, Department of Mathematics and Computer Science	August 2025-
	University of California, Irvine , Irvine, CA, USA Visiting Assistant Professor, Department of Mathematics	July 2024-July 2025
	Vanderbilt University , Nashville, TN, USA Postdoc Scholar (Research), Department of Mathematics Mentor: Dr. Gieri Simonett	Sep. 2021-June 2024
EDUCATION	Purdue University , West Lafayette, IN Ph.D., Mathematics Advisor: Dr. Changyou Wang	Aug. 2016-Aug. 2021
	Beijing Normal University , Beijing, China M.S., Computational Mathematics Advisor: Dr. Jiequan Li	Sep. 2013-July 2016
	South China Normal University , Guangzhou, China B.S., Mathematics and Applied Mathematics	Sep. 2009-July 2013

PUBLICATIONS

Diffusion and Optimal Transport in Machine Learning

1. Xinyang Liu, Hengrong Du, Wei Deng, Ruqi Zhang. *Optimal Stochastic Trace Estimation in Generative Modeling* . AISTATS 2025.
2. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Linear Partial Gromov-Wasserstein Embedding* . ICLR 2025.
3. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Partial Gromov-Wasserstein Metric* . ICLR 2025.
4. Haoyang Zheng, Hengrong Du, Qi Feng, Wei Deng, Guang Lin. *Constrained exploration via reflected replica exchange stochastic gradient langevin dynamics*. ICML 2024.
5. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *Reflected Schrödinger bridge for constrained generative modeling*. UAI 2024.

6. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *On convergence of approximate Schrödinger bridge with bounded cost*. ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems (2023).

Complex Fluids

1. Hengrong Du, Gieri Simonett, Yuanzhen Shao. *On a thermodynamically consistent model for magnetoviscoelastic fluids in 3D*. J. Evol. Equ. 24 (2024), no. 9, 1-51.
2. Hengrong Du, Yuanzhen Shao, Gieri Simonett. *Well-posedness for magneto-viscoelastic fluids in 3D*. Nonlinear Anal. Real World Appl. 69 (2023), no. 103759.
3. Hengrong Du, Tao Huang, Changyou Wang. *Weak compactness property of simplified nematic liquid crystal flows in dimension two*. Math. Z. 302 (2022), no. 2, 2111-2130.
4. Hengrong Du, Changyou Wang. *Global weak solutions to the stochastic Ericksen–Leslie system in dimension two*. Discrete Contin. Dyn. Syst. 42 (2022), no. 5, 2175-2197.
5. Hengrong Du, Changyou Wang. *Partial regularity of a nematic liquid crystal model with kinematic transport effects*. Nonlinearity 34 (2021), no. 5, 3001-3045.
6. Hengrong Du, Yimei Li, Changyou Wang. *Weak solutions of non-isothermal nematic liquid crystal flow in dimension three*. J. Elliptic Parabol. Equ. 6 (2020), no. 1, 71-98.
7. Hengrong Du, Xianpeng Hu, Changyou Wang. *Suitable Weak Solutions for the Co-rotational Beris–Edwards System in Dimension Three*. Arch. Ration. Mech. Anal. 238 (2020), no. 2, 749-803.

Calculus of Variations

1. Hengrong Du, Nung Kwan Yip. *Stability of self-similar solutions to geometric flows*. Interfaces Free Bound. 25 (2023), no. 2, 155–191.
2. Hengrong Du, Qinfeng Li, Changyou Wang. *Compactness of M -uniform domains and optimal thermal insulation problems*. Adv. Calc. Var. 16 (2023), no. 1, 17-43.

PREPRINTS

1. Hengrong Du, Qi Feng, Changwei Tu, Xiaoyu Wang, Lingjiong Zhu. *Non-Reversible Langevin Algorithms for Constrained Sampling*. [arXiv:2501.11743](#).
2. Yikun Bai, Huy Tran, Hengrong Du, Xinran Liu, Soheil Kolouri. *Fused Partial Gromov-Wasserstein for Structured Objects*. [arXiv:2401.03662](#).
3. Hengrong Du, Chuntian Wang. *Partial regularity for the three-dimensional stochastic Ericksen–Leslie Equations*. [arXiv:2401.03662](#).

AWARDS

- Bilsland Dissertation Fellowship June 2020-May 2021
It was intended to give the most accomplished final-year PhD candidates an opportunity to complete the dissertation within the 2020–21 academic year by devoting full-time effort to research and writing.

TALKS

- PDE & Applied Mathematics seminar
University of California, Riverside January 2025
- PDEs & Applications Seminar
Queen's University January 2025
- Applied Math Seminar
The University of Alabama October 2024
- PDEs and Fluid Mechanics and Atmospheric Sciences
West Virginia University April 2024
- AMS Special Session on Fluids: Analysis, Applications, and Beyond
Florida State University March 2024
- Financial Mathematics Seminars
Florida State University March 2024
- AMS Special Section on Dynamics and Regularity of PDEs
Joint Mathematics Meeting January 2024
- Seminar on Analysis and Stochastic Analysis
Auburn University November 2023
- PDE Seminar
University of Tennessee, Knoxville October 2023
- PDE/Applied Math Seminar
Indiana University, Bloomington October 2023
- Richard F. Barry Jr. Seminar Series
Old Dominion University September 2023
- The 13th AIMS Conference
University of North Carolina Wilmington June 2023
- AMS Spring Central Sectional Meeting
University of Cincinnati April 2023
- Undergraduate Seminar in Mathematics
Vanderbilt University March 2023
- Analysis Seminar
Wayne State University March 2023

	• 3rd Biennial Meeting of SIAM Pacific Northwest Section Washington State University May 2022 • AMS Spring Central Virtual Sectional Meeting Purdue University March 2022 • Graduate Online Mini-course Beijing Normal University November 2021 • PDE Seminar Fall 2021 Vanderbilt University September 2021 • Analysis Seminar The University of Alabama February 2021 • PDE Seminar Purdue University September 2020 • Student Analysis Seminar Purdue University February 2020 • Student Analysis Seminar Purdue University February 2020 • PDE Seminar Purdue University January 2020
CONFERENCE PARTICIPATION	• Rivière-Fabes Symposium on Analysis and PDE University of Minnesota Apr. 19-21, 2023 • Shanks Workshop on Advances in Mathematical and Theoretical Biology Vanderbilt University Mar. 17-19, 2023 • Workshop on Geometry and Analysis of Fluid Flows Stony Brook University Jan. 16-20, 2023 • AMS Fall Central Sectional Meeting University of Texas at El Paso Sep. 17-18, 2022 • Shanks Workshop on Mathematical Aspect of Fluid Dynamics Vanderbilt University Feb. 19-20, 2022 • KUMUNU-ISU Conference on PDE, Dynamical Systems, and Applications University of Nebraska-Lincoln Oct. 23-24, 2021 • AMS Fall Western Sectional Meeting Online Oct. 23-24, 2021 • The 84th Midwest PDE Seminar Illinois Institute of Technology Oct. 26-27, 2019 • Midwest Geometry Conference 2019 Iowa State University Sept. 6-8, 2019 • The 83rd Midwest PDE Seminar Indiana University Bloomington Mar. 30-31, 2019
TEACHING EXPERIENCE	Instructor at University of California, Irvine Fall 2024-Present • MATH 3A Introduction to Linear Algebra Spring 2025 • MATH 112B Introduction to Partial Differential Equation Winter 2025 • MATH 3D Elementary Differential Equations Fall 2024

Instructor at Vanderbilt University	Fall 2021-Present
• MATH 3100 Introduction to Analysis	Spring 2024
• MATH 2400 Differential Equations with Linear Algebra	Fall 2023
• MATH 3100 Introduction to Analysis	Spring 2023
• MATH 3100 Introduction to Analysis	Fall 2022
• MATH 1301 Accelerated Single-Variable Calculus II	Fall 2022
• MATH 2420 Methods of Ordinary Differential Equations	Spring 2022
• MATH 2400 Differential Equations with Linear Algebra	Spring 2022
Recitation Instructor at Purdue University	Fall 2016-Summer 2021
• MA 26200 Linear Algebra And Differential Equations	Fall 2019
• MA 16200 Plane Analytic Geometry And Calculus II	Spring 2019
• MA 16500 Plane Analytic Geometry And Calculus I	Fall 2018